ASMAE EL HARTI

General Engineer – Mechatronics & Cyber-Physical Systems

Ifrane, Morocco • a.elharti@aui.ma • github.com/addy79 • linkedin.com/in/asmae-el-harti-9b76b2268

EDUCATION

- **Al Akhawayn University in Ifrane (AUI)** — *Ifrane, Morocco* (Expected May 2027)
  Bachelor of Science in General Engineering (BSGE) — Concentration: **Mechatronics & Robotics**
  Relevant Coursework: Kinematics & Dynamics of Mechanisms, Control Systems, Microcontroller Interfacing, Mechanics of Materials, Fluid Mechanics, Electronic Circuits, Computer-Aided Design.
- **ALX Africa** — *Remote* (2023 – Present)
  Software Engineering Specialization — Low-level systems programming in C, Linux OS internals, memory management, socket networking, concurrent architectures, Python backend pipelines.

TECHNICAL COMPETENCIES

- **Mechanical Design & CAD:** FreeCAD (author of 44 parametric mechanisms), SolidWorks, Cycloidal Transmissions, Harmonic Drives, Planetary Gearboxes, Stewart Hexapod Platforms, Delta Robots, Involute & Epicyclic Gearing, 3D Printing (FDM/Resin), Press-fits & Tolerance Analysis.
- **Embedded Systems & Hardware:** Raspberry Pi Zero, Arduino, STM32, C/C++, Python, Low-Latency UDP Bridges, UART, SPI, I2C, Hardware-in-the-Loop (HIL) Telemetry, Power Distribution, Oscilloscopes, Bench Diagnostics.
- **Spatial Computing & Simulation:** Godot 4, OpenXR Virtual Reality, Multi-Body Dynamics, GPU Shaders, Real-Time Sensor Synchronization, Networked Multi-User Simulation.
- **Controls & Kinematics:** Forward & Inverse Kinematics (FK/IK), PID Regulation, Trajectory Planning, State Estimation, BLDC 3-Phase Motor Control, Dynamometer Testing, Back-EMF Waveform Analysis.

FLAGSHIP ENGINEERING PROJECTS

- **Digitwin1 – Cyber-Physical Robotic Actuator Digital Twin** (Godot 4, OpenXR, Raspberry Pi Zero, UDP, Python)
  Architected an end-to-end cyber-physical digital twin connecting a 1-DOF physical actuator with an OpenXR spatial simulation. Built a sub-millisecond UDP bridge daemon streaming real-time encoder angles, torque setpoints, dynamic joint limits, and hardware safety interlocks.
- **FreeCAD Engineering Mechanism Suite** (44 Fully Parametric Mechanical Assemblies)
  Engineered from first principles 44 complex mechanisms: zero-backlash cycloidal reducers, harmonic strain-wave drives, 6-axis Stewart platform, 3-axis Delta robot, 6-DOF industrial robot arm, and dual-spool jet engine cutaway. Built with fully parametric spreadsheet constraints.
- **VRMechanics & BrushlessLab – OpenXR Engineering Laboratories** (Godot 4, OpenXR, Mechatronics)
  Developed interactive spatial laboratories for mechanical teardowns of multi-body systems and 3-phase BLDC actuator diagnostics (milliohm winding resistance, back-EMF waveform scope, Hall sensor timing, bearing envelope wear).
- **Pioneer ROV – Marine Robotics Submersible** (AUI Robotics Club, Electronics & Telemetry Lead)
  Designed the electrical power distribution system, 6-thruster vector control allocation matrix, and tethered RS-485/Ethernet sensor telemetry for the underwater exploration robot competing in regional marine challenges.

LANGUAGES

- **Arabic:** Native • **English:** Fluent / Academic Working Proficiency • **French:** Bilingual / Professional Proficiency